

Chapter 2

Groups, First Encounter

2.1 Definition of a Group

Exercise 2.1.1. Write a careful proof that every group is the group of isomorphisms of a groupoid. In particular, every group is the group of automorphisms of some object in some category.

Solution. Let G be any group. It suffices to construct such a groupoid such that the collection of isomorphisms is isomorphic to G . Let $\text{Obj}(\mathbf{G}) = \{*\}$. For all $g \in G$, let there exist a morphism $f_g \in \text{End}_{\mathbf{G}}(*)$, and let morphism composition be defined as

$$f_g \circ f_h = f_{gh},$$

for $g, h \in G$, using the notation as previously. To show this is a category, it suffices to show there is an identity morphism. Note that f_e is clearly an identity, since $f_g \circ f_e = f_e \circ f_g = f_g$. Therefore, $\mathbf{G}$ is a category. Now we show $\mathbf{G}$ is a groupoid. Let $f_g \in \text{Hom}_{\mathbf{G}}(*, *)$. Since $g \in G$, there exists $g^{-1} \in G$ such that $gg^{-1} = g^{-1}g = e$. Equivalently, there exists $f_{g^{-1}} \in \text{End}_{\mathbf{G}}(*)$ such that

$$f_g \circ f_{g^{-1}} = f_{g^{-1}} \circ f_g = f_e.$$

Therefore, all morphisms have inverses, and $\mathbf{G}$ is a groupoid. Since $\mathbf{G}$ has only one element, the collection of (iso)morphisms in $\mathbf{G}$ is precisely $\text{Aut}_{\mathbf{G}}(*)$, which we have shown to be a group already. Let $\phi: G \rightarrow \text{Aut}_{\mathbf{G}}(*)$ be the map $g \mapsto f_g$. Clearly the map is well-defined and surjective. Also note that

$$\phi(gh) = f_{gh} = f_g \circ f_h = \phi(g)\phi(h).$$

Therefore, ϕ is a homomorphism, and we need only show ϕ is injective. Suppose $\phi(g) = \phi(h)$. That is, $f_g = f_h$. Since inverses are unique, there exists exactly one inverse f_a to the above morphisms. Since the identity is unique, we must have $ag = ga = e$ and $ah = ha = e$. Cancelling a , we have $g = h$. ■

Exercise 2.1.2. Consider the sets $\emptyset, \mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ under $+$ and $\cdot$. Which are groups, and which aren't?

Solution. Clearly $\mathbb{N}$ is a group under none since it isn't closed under addition and has no multiplicative inverses. $\mathbb{Z}$ is a group under addition but not under multiplication. $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ are all groups under $+$, and are groups under $\cdot$ if zero is removed. ■

Exercise 2.1.3. Prove that $(gh)^{-1} = h^{-1}g^{-1}$ for all $g, h \in G$ a group.

Solution. Let $g, h \in G$. Note that

$$(gh)(h^{-1}g^{-1}) = g(hh^{-1})g^{-1} = ge_Gg^{-1} = gg^{-1} = e_G.$$

Therefore, $(gh)^{-1} = h^{-1}g^{-1}$. ■

Exercise 2.1.4. Suppose $g^2 = e$ for all $g \in G$ a group. Prove G is abelian.

Solution. Let $g, h \in G$. Since $g^2 = e$ for all $g \in G$, we have $g = g^{-1}$ for all $g \in G$. By the previous exercise, $(gh)^{-1} = h^{-1}g^{-1}$. However, since $g = g^{-1}$ for all $g \in G$, we have

$$gh = hg.$$

■

Exercise 2.1.5. Prove every row and column of the Cayley table for a group contains each element exactly once.

Solution. It suffices to show that left- and right-multiplication by a group element is a bijection $G \rightarrow G$. Let $g \in G$, and consider the map $a \mapsto ga$. Clearly this map is well defined, and it has an inverse map $a \mapsto g^{-1}a$, so it is a bijection. Right multiplication follows by the same logic. ■

Exercise 2.1.6. Prove there is only one multiplication table for groups with 1, 2, or 3 elements. Show there are two tables for groups with 4 elements.

Solution. The case for 1 is trivial. For the second case, let $\{e, g\}$ be a group. We must have $eg = ge = e$, $ee = e$, and since g needs an inverse, we must have $gg = e$. Therefore, the group requirement forces this unique multiplication structure.

Now let $\{e, a, b\}$ be a group. Suppose for contradiction that a, b are not inverses. Then ab is either a or b . The inverse of a must then be a , and likewise for b . It follows that $e = a^2 = aba$. Cancelling a once on the left and right gives us $a^2 = e = b$, a contradiction. Therefore, $a = b^{-1}$, and this forces a unique structure.

I have done this proof before and I don't feel like repeating it so I'm just going to leave it here. ■

Fuck this

2.2 Examples of Groups

2.3 The Category Grp

Exercise 2.3.1. Let $\phi : G \rightarrow H$ be a morphism in a category $\mathcal{C}$ with products. Explain why there is a unique morphism $(\phi \times \phi) : G \times G \rightarrow H \times H$ compatible in the evident way with the natural projections.

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Solution. I think it's good to spell out what the “evident way” is. We want the diagram

$$\begin{array}{ccccc}
 & & G & \xrightarrow{\phi} & H \\
 & \nearrow \pi & & \nearrow \pi & \\
 G \times G & \xrightarrow{\phi \times \phi} & H \times H & & \\
 & \searrow \pi & & \searrow \pi & \\
 & & G & \xrightarrow{\phi} & H
 \end{array}$$

to commute. Note that ϕ determines a map $G \times G \rightarrow H$ given by $\phi\pi$. By the universal property of products, there exists precisely one morphism $\sigma : G \times G \rightarrow H \times H$ such that the diagram

$$\begin{array}{ccc}
 & \xrightarrow{\phi\pi} & H \\
 & \nearrow \pi & \\
 G \times G & \xrightarrow{\sigma} & H \times H \\
 & \searrow \pi & \\
 & \xrightarrow{\phi\pi} & H
 \end{array}$$

commutes. Note that these diagrams are the same, except $\phi \times \phi$ has been replaced by σ . We can take this unique morphism σ to be the morphism $\phi \times \phi$. ■

Exercise 2.3.2. Let $\phi : G \rightarrow H$, $\psi : H \rightarrow K$ be morphisms in a category $\mathbf{C}$ with products, and consider morphisms between the products $G \times G$, $H \times H$, and $K \times K$. Prove that

$$(\psi\phi) \times (\psi\phi) = (\psi \times \psi)(\phi \times \phi).$$

Solution. For simplicity, since the diagrams are symmetric, I will only draw the one side of them. We have

$$\begin{array}{ccccc}
 G \times G & \xrightarrow{\phi \times \phi} & H \times H & \xrightarrow{\psi \times \psi} & K \times K \\
 \downarrow \pi & & \downarrow \pi & & \downarrow \pi \\
 G & \xrightarrow{\phi} & H & \xrightarrow{\psi} & K
 \end{array}$$

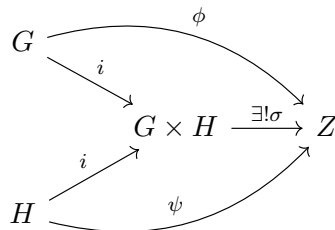
Note that by previous logic, $(\psi \times \psi)(\phi \times \phi)$ is a unique morphism $G \times G \rightarrow K \times K$. It suffices to show then that $(\psi\phi) \times (\psi\phi)$ exists, and is a morphism $G \times G \rightarrow K \times K$. This follows fairly immediately. Consider the commutative diagram

$$\begin{array}{ccc}
 G \times G & \xrightarrow{\exists! \sigma} & K \times K \\
 \downarrow \pi & & \downarrow \pi \\
 G & \xrightarrow{\psi\phi} & K
 \end{array}$$

Since the products are symmetrical, this is in fact the full diagram for the universal property of products. By the same logic then, we have $\sigma = (\psi\phi) \times (\psi\phi)$. ■

Exercise 2.3.3. Show that if G, H are abelian groups, then $G \times H$ satisfies the universal property for coproducts in $\mathbf{Ab}$.

Solution. Recall the universal property of coproducts



yeah idfk why it has to be abelian ngl

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